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April 14, 2026

To:

Program Officers: José Guzmán, PhD, Megan Ewing, Sarah Tanja, Andy Nutzhorn
Funding Organization: International Society of Crustacean Ecophysiology
Program Title: *Applied Comparative Ecophysiology of Hemigrapsus oregonensis and Carcinus maenas: Linking Mechanisms to Management in a Changing Coastal Environment*

Subject: Letter of Intent to Submit a Proposal – *Applied Comparative Ecophysiology of Hemigrapsus oregonensis and Carcinus maenas*

Dear José Guzmán,

I am writing to formally express my intent to submit a proposal in response to the RFP titled “*Applied Comparative Ecophysiology of Hemigrapsus oregonensis and Carcinus maenas: Linking Mechanisms to Management in a Changing Coastal Environment.*”

Our proposed project, tentatively titled “**How does Puget Sound Pollutants affect *Hemigrapsus oregonensis*’ response to predators?**”, will address the following thematic area(s) of the RFP:

- Pollutants effects on chemosensory physiology
- Applied Stress Physiology of Marine Invertebrates
- Invasive species predation avoidance

This project will investigate how common pollutants found in the Puget Sound affect the stress response from predator cues in *H. oregonensis*. By exploring this question through this project, we will be able to see if and how these pollutants may prevent predator detection by using stress response as a proxy. This will shed light on how human pollutants have affected native shore crab’s ability to avoid predation from an increasingly invasive population of European green crabs *C. maenas*.

The anticipated Principal Investigators will be Noah Ozguner, Callum Rangan, Abigail Kwan and Jack Brown. Our research approach emphasizes ecologically meaningful physiological endpoints with direct application to coastal resource management and climate adaptation.

We appreciate this opportunity to contribute to applied coastal science and would be happy to provide any additional information if needed prior to full proposal submission.

Sincerely,

Jack Brown, Noah Ozguner, Callum Rangan, Abigail Kwan